Variable Selection for Wasserstein Robust Hypothesis Testing

Abstract

Starting from the finite least-favorable-distribution formulation of Wasserstein robust testing, we derive a convex model with coordinate selection. A uniform risk bound introduces the selection variables without changing binary optimal values. The resulting convex value function admits affine dual cuts for optimization over coordinate subsets.

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we follow the convention in [2] to define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \leq \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \leq d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Finite reformulation

Assume $n_0, n_1 \geq 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. All distributions have finite first moments. Fix $S \subseteq [D]$ and write

$$V(S) = \min_{f \in \mathcal{F}_S} R(f), \qquad \mathcal{X} = \{x_i^{(0)} : i \in [n_0]\} \cup \{x_j^{(1)} : j \in [n_1]\}.$$

Sources retain their class indices; a target is denoted by $y \in \mathcal{X}$. Projection onto S does not increase the transport cost. Conversely, any transport of projected observations lifts to the original space by keeping each source's unselected coordinates unchanged, with the same cost. Thus the projected Wasserstein balls describe exactly the uncertainty relevant to a detector $g(x[S])$.

Following [2], define

$$\psi(p) = \inf_{v \in \mathbb{R}} \{p\ell(v) + (1 - p)\ell(-v)\}, \qquad 0 \leq p \leq 1.$$

The following is the finite LFD formulation of [2, Proposition 1 and Theorem 3, equation (6)], applied to the projected observations and written with their original class indices. A summand with zero total mass is defined as zero.

Proposition 2.1 (Finite LFD representation)**.** *The fixed-selection risk is*

$$
\begin{aligned}
V(S) = \max_{p,\gamma\geq 0} \quad & \sum_{y\in\mathcal{X}} (p_0(y)+p_1(y))\psi\left(\frac{p_0(y)}{p_0(y)+p_1(y)}\right) \\
\textit{subject to} \quad & \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, && i\in[n_k],\ k=0,1, \\
& p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}, && y\in\mathcal{X},\ k=0,1, \\
& \sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S]-y[S]\|_1 \leq \rho_k, && k=0,1.
\end{aligned}
\tag{3}
$$

Here $\gamma_{i,y}^{(k)}$ is the mass moved from $x_i^{(k)}[S]$ to $y[S]$, and $p_k(y)$ is its resulting target mass. All targets are retained, including the source itself. If projected targets coincide, their masses are aggregated. Concavity and positive homogeneity of the objective imply that aggregation cannot decrease it; conversely, an aggregate can be split in the same proportions for both classes without changing the objective. Hence the fixed target labels $y\in\mathcal{X}$ do not affect the value.

Write g_y for the score assigned to $y[S]$. Using $v=-g_y$ and substituting the target masses gives

$$
\sum_{y\in\mathcal{X}} (p_0(y)+p_1(y))\psi\left(\frac{p_0(y)}{p_0(y)+p_1(y)}\right) = \inf_g \left\{ \sum_{i,y} \gamma_{i,y}^{(0)} \ell(-g_y) + \sum_{j,y} \gamma_{j,y}^{(1)} \ell(g_y) \right\}.
$$

The transport set is compact and convex, and the expression inside braces is convex in the finite score vector g and linear in γ. Finite-dimensional minimax therefore gives

$$
V(S) = \inf_g \left\{ \max_{\gamma^{(0)}} \sum_{i,y} \gamma_{i,y}^{(0)} \ell(-g_y) + \max_{\gamma^{(1)}} \sum_{j,y} \gamma_{j,y}^{(1)} \ell(g_y) \right\},
$$

where each maximum satisfies its source-mass and transport-budget constraints in (3).

For fixed g, the class-0 maximization is a transport LP. Assign a free multiplier $a_i^{(0)}$ to source-mass conservation and a nonnegative multiplier λ_0 to the transport budget. Its Lagrangian is

$$
\begin{aligned}
\mathcal{L}_0 = & \sum_{i,y} \gamma_{i,y}^{(0)} \ell(-g_y) + \sum_i a_i^{(0)} \left(\frac{1}{n_0} - \sum_y \gamma_{i,y}^{(0)} \right) \\
& + \lambda_0 \left(\rho_0 - \sum_{i,y} \gamma_{i,y}^{(0)} \|x_i^{(0)}[S]-y[S]\|_1 \right) \\
= & \frac{1}{n_0}\sum_i a_i^{(0)} + \rho_0\lambda_0 \\
& + \sum_{i,y} \gamma_{i,y}^{(0)} \left[\ell(-g_y) - a_i^{(0)} - \lambda_0 \|x_i^{(0)}[S]-y[S]\|_1 \right].
\end{aligned}
\tag{4}
$$

The supremum over $\gamma^{(0)}\geq 0$ is finite exactly when each bracketed coefficient is nonpositive. Applying LP duality to both classes, then minimizing over g, yields

$$
\begin{aligned}
V(S) = \min_{g,a,\lambda} \quad & \frac{1}{n_0}\sum_i a_i^{(0)} + \frac{1}{n_1}\sum_j a_j^{(1)} + \rho_0\lambda_0 + \rho_1\lambda_1 \\
\text{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \lambda_0 \|x_i^{(0)}[S]-y[S]\|_1, && i\in[n_0],\ y\in\mathcal{X}, \\
& \ell(g_y) \leq a_j^{(1)} + \lambda_1 \|x_j^{(1)}[S]-y[S]\|_1, && j\in[n_1],\ y\in\mathcal{X}, \\
& g\in\mathbb{R}^{\mathcal{X}}, \quad a^{(k)}\in\mathbb{R}^{n_k}, \quad \lambda_k\geq 0, && k=0,1.
\end{aligned}
\tag{5}
$$

Denote this objective by $J(a, \lambda)$. Scores at coincident projections can be chosen equal because their constraint columns are identical. Appendix A extends these finite scores to the measurable detector g in Section 1 and verifies attainment of the minimum.

The zero detector has risk 2. Taking $y = x_i^{(k)}$ in the constraints gives $a_i^{(k)} \geq 0$, so an optimum satisfies

$$\rho_0\lambda_0 + \rho_1\lambda_1 \leq J(a, \lambda) = V(S) \leq 2. \tag{6}$$

This bound will introduce coordinate selection without changing the binary optimal values.

3 Selection variables and duality

Let $z = \mathbf{1}_S$ and $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \leq d\}$. The transport cost decomposes directly into coordinates:

$$\|x_i^{(k)}[S] - y[S]\|_1 = \sum_{m=1}^{D} z_m |x_i^{(k)}[m] - y[m]|.$$

Introduce w_{km} in place of $\lambda_k z_m$. The conditions $0 \leq w_{km} \leq \lambda_k$ and $\rho_0 w_{0m} + \rho_1 w_{1m} \leq 2z_m$ force both weights to zero when a coordinate is unselected. When it is selected, setting $w_{km} = \lambda_k$ is allowed at an optimum by (6). For $z \in [0,1]^D$, define

$$\begin{aligned}
F(z) = \min_{g,a,\lambda,w} \quad & J(a, \lambda) \\
\text{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \sum_m w_{0m} |x_i^{(0)}[m] - y[m]|, && i \in [n_0],\ y \in \mathcal{X}, \\
& \ell(g_y) \leq a_j^{(1)} + \sum_m w_{1m} |x_j^{(1)}[m] - y[m]|, && j \in [n_1],\ y \in \mathcal{X}, \\
& 0 \leq w_{km} \leq \lambda_k, && k = 0,1,\ m \in [D], \\
& \rho_0 w_{0m} + \rho_1 w_{1m} \leq 2z_m, && m \in [D], \\
& g \in \mathbb{R}^{\mathcal{X}}, \quad a^{(k)} \in \mathbb{R}^{n_k}, \quad \lambda_k \geq 0, && k = 0,1.
\end{aligned} \tag{7}$$

This has the logical structure of problems (2.1)–(2.2) in [1]: the outer problem selects coordinates, each activating a pair (w_{0m}, w_{1m}) in a convex inner problem.

Proposition 3.1 (Selection formulation)**.** *The minimum in* (7) *is attained. The function F is convex on $[0,1]^D$, and*

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S| \leq d} V(S) = \min_{z \in \mathcal{Z}_d} F(z). \tag{8}$$

Proof. An optimum of (5) extends to (7) by $w_{km} = \lambda_k z_m$ at $z = \mathbf{1}_S$, using (6). Conversely, a feasible point at that binary selection has $w_{km} = 0$ outside S, hence

$$\sum_m w_{km} |x_i^{(k)}[m] - y[m]| \leq \lambda_k \sum_{m \in S} |x_i^{(k)}[m] - y[m]| = \lambda_k \|x_i^{(k)}[S] - y[S]\|_1.$$

It is therefore feasible in (5) with the same objective. This proves equality of the binary optimal values. For any z, the feasible choice $g = 0$, $a = \mathbf{1}$, $\lambda = w = 0$ has value 2. The sublevel $J \leq 2$ is compact by the bounds in Appendix A and $0 \leq w_{km} \leq \lambda_k$. All constraints are jointly convex, including in z. Taking convex combinations of feasible solutions gives

$$F(tz + (1-t)z') \leq tF(z) + (1-t)F(z'), \qquad 0 \leq t \leq 1.$$

□

3.1 Dual representation

Assign multipliers $\gamma_{i,y}^{(0)} \geq 0$ and $\gamma_{j,y}^{(1)} \geq 0$ to the two families of loss constraints in (7), and $\mu_m \geq 0$ to each coordinate-capacity constraint. Minimizing over the free source variables a recovers the source-mass equalities $\sum_y \gamma_{i,y}^{(k)} = 1/n_k$. Thus the same transport notation reappears through duality. With

$p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}$ as in (3), put

$$\begin{aligned} H(\gamma) &= \sum_{y \in \mathcal{X}} \inf_{v \in \mathbb{R}} \{p_0(y)\ell(-v) + p_1(y)\ell(v)\} \\ &= \sum_{y \in \mathcal{X}} (p_0(y) + p_1(y))\psi\left(\frac{p_0(y)}{p_0(y) + p_1(y)}\right). \end{aligned} \tag{9}$$

This is the LFD objective, obtained by minimizing over the target scores. The remaining dual constraints define the set

$$\mathcal{D} = \left\{ (\gamma, \mu) \geq 0 : \begin{array}{ll} \displaystyle\sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = 1/n_k, & i \in [n_k],\ k = 0, 1, \\ \displaystyle\sum_{m=1}^{D} \left(\sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} |x_i^{(k)}[m] - y[m]| - \rho_k \mu_m \right)_+ \leq \rho_k, & k = 0, 1 \end{array} \right\}, \tag{10}$$

where $(v)_+ = \max\{v, 0\}$. Appendix B derives the second line by minimizing over λ, w. In particular, $\mathcal{D}$ does not depend on z.

Proposition 3.2 (Dual representation)**.** *For every $z \in [0,1]^D$, the following maximum is attained:*

$$F(z) = \max_{(\gamma,\mu) \in \mathcal{D}} \{H(\gamma) - 2\mu^\top z\}. \tag{11}$$

An optimizer (γ^t, μ^t) at z^t gives the supporting function

$$c_t(z) = H(\gamma^t) - 2(\mu^t)^\top z = F(z^t) - 2(\mu^t)^\top (z - z^t) \leq F(z), \tag{12}$$

with $c_t(z^t) = F(z^t)$.

The function H is concave as an infimum of linear functions, and $\mathcal{D}$ is convex. The dual at fixed z is therefore a convex optimization problem. Its affine dependence on z gives the cutting-plane construction underlying Theorem 2.5 and Algorithm 3.1 of [1].

4 Cutting-plane algorithm

Write $\mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z)$. After evaluating $z^1, \ldots, z^t$ and obtaining their dual optimizers, let $U_t = \min_{s \leq t} F(z^s)$ and solve

$$\begin{aligned} (z^{t+1}, L_t) \in \operatorname*{arg\,min}_{z \in \mathcal{Z}_d,\, \eta \geq 0} \quad & \eta \\ \text{subject to} \quad & \eta \geq F(z^s) - 2(\mu^s)^\top (z - z^s), \qquad s = 1, \ldots, t. \end{aligned} \tag{13}$$

Every retained cut is a global lower bound and is exact at its evaluated selection. Consequently,

$$L_t \leq \mathrm{OPT} \leq U_t. \tag{14}$$

Starting from any $z^1 \in \mathcal{Z}_d$, alternate the convex subproblem (7)–(11) and the binary linear master (13). Stop when $U_t - L_t \leq \varepsilon$ and return the best evaluated selection. Appendix A recovers its detector, with risk U_t and optimality gap at most ε. For an inexact inner solve, use the actual feasible dual expression $H(\gamma^t) - 2(\mu^t)^\top z$ as the cut and a primal feasible objective as the upper bound; the master must provide a global lower bound. The binary formulation is exact, although its continuous relaxation can be weak.

A Recovery of a detector

For any feasible (g, a, λ) in (5), choose equal scores at coincident projected targets. This preserves feasibility and the objective. We construct a measurable extension satisfying

$$\begin{aligned} \ell(-g(\xi)) \leq E_0(\xi), \qquad & \ell(g(\xi)) \leq E_1(\xi), \\ E_k(\xi) = \min_{i \in [n_k]} \{a_i^{(k)} + \lambda_k \|x_i^{(k)}[S] - \xi\|_1\}, \qquad & k = 0, 1. \end{aligned} \tag{15}$$

Proof. At an observed projection, set $g(y[S]) = g_y$; the constraints of (5) give (15) there. Elsewhere, choose the smallest minimizing indices i, j in $E_0(\xi), E_1(\xi)$, and set

$$r_0 = \|x_i^{(0)}[S] - \xi\|_1, \qquad r_1 = \|x_j^{(1)}[S] - \xi\|_1, \qquad b = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1.$$

For $b > 0$, let $\tau = \min\{1, r_0/b\}$ and define

$$g(\xi) = (1-\tau)g_{x_i^{(0)}} + \tau g_{x_j^{(1)}}. \tag{16}$$

The triangle inequality gives $\tau b \le r_0$ and $(1-\tau)b \le r_1$. Convexity and (5) imply

$$\ell(-g(\xi)) \le (1-\tau)\ell(-g_{x_i^{(0)}}) + \tau\ell(-g_{x_j^{(1)}}) \le a_i^{(0)} + \lambda_0\tau b \le E_0(\xi),$$
$$\ell(g(\xi)) \le (1-\tau)\ell(g_{x_i^{(0)}}) + \tau\ell(g_{x_j^{(1)}}) \le a_j^{(1)} + \lambda_1(1-\tau)b \le E_1(\xi).$$

For $b = 0$, take $g(\xi) = g_{x_i^{(0)}}$; the same bounds follow from (5). The finite tie rule makes g Borel measurable, and $|g| \le \max_{y\in\mathcal{X}} |g_y|$. For any $\mathbb{P}_k \in \mathcal{P}_k$, integrate (15) against an optimal empirical-to-$\mathbb{P}_k$ coupling, using that projection cannot increase its cost. This gives

$$\mathbb{E}_{\mathbb{P}_0}\ell(-g(X[S])) \le \frac{1}{n_0}\sum_i a_i^{(0)} + \rho_0\lambda_0,$$
$$\mathbb{E}_{\mathbb{P}_1}\ell(g(X[S])) \le \frac{1}{n_1}\sum_j a_j^{(1)} + \rho_1\lambda_1.$$

Thus $f(x) = g(x[S])$ satisfies $R(f) \le J(a,\lambda)$. On the nonempty sublevel $J \le 2$, $0 \le a_i^{(k)} \le 2n_k$ and $0 \le \lambda_k \le 2/\rho_k$. Both losses of every g_y are bounded by the two classes' constraints. Since $\ell(t) \to \infty$ as $t \to \infty$, all scores are bounded. This sublevel is compact, so (5) attains its minimum; the extension of an optimum has risk $V(S)$. □

B Proof of the dual representation

Proof of Proposition 3.2. In addition to γ, μ, assign multipliers $\beta_{km} \ge 0$ to $w_{km} \le \lambda_k$. Keeping g, a free and $\lambda, w \ge 0$, the minimization Lagrangian is

$$\begin{aligned}\mathcal{L} = J(a,\lambda) &+ \sum_{i,y}\gamma_{i,y}^{(0)}\left[\ell(-g_y) - a_i^{(0)} - \sum_m w_{0m}|x_i^{(0)}[m] - y[m]|\right] \\ &+ \sum_{j,y}\gamma_{j,y}^{(1)}\left[\ell(g_y) - a_j^{(1)} - \sum_m w_{1m}|x_j^{(1)}[m] - y[m]|\right] \\ &+ \sum_{k,m}\beta_{km}(w_{km} - \lambda_k) + \sum_m \mu_m(\rho_0 w_{0m} + \rho_1 w_{1m} - 2z_m).\end{aligned} \tag{17}$$

The infimum over a is finite exactly when $\sum_y \gamma_{i,y}^{(k)} = 1/n_k$. The terms depending on g are $\sum_y [p_0(y)\ell(-g_y) + p_1(y)\ell(g_y)]$; their infimum is $H(\gamma)$. The coefficients of $\lambda, w \ge 0$ must be nonnegative:

$$\rho_k - \sum_m \beta_{km} \ge 0, \qquad \beta_{km} + \rho_k\mu_m - \sum_{i,y}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]| \ge 0.$$

Together with $\beta \ge 0$, these are equivalent to

$$\sum_m \left(\sum_{i,y}\gamma_{i,y}^{(k)}|x_i^{(k)}[m] - y[m]| - \rho_k\mu_m\right)_+ \le \rho_k.$$

Eliminating β therefore leaves the domain $\mathcal{D}$ and objective $H(\gamma) - 2\mu^\top z$.

For every fixed $z \in [0,1]^D$, the point $g = 0$, $a = 2\mathbf{1}$, $\lambda = \mathbf{1}$, $w = 0$ strictly satisfies all loss inequalities and satisfies all affine constraints. Treating nonnegativity as affine constraints, the refined Slater condition

gives strong duality and dual attainment, including at boundary selections. Finally, any dual feasible point gives a global lower bound. Optimality of (γ^t, μ^t) at z^t gives

$$F(z) \geq H(\gamma^t) - 2(\mu^t)^\top z = F(z^t) - 2(\mu^t)^\top (z - z^t),$$

which proves (12). □

References

[1] Dimitris Bertsimas, Ryan Cory-Wright, and Jean Pauphilet. A unified approach to mixed-integer optimization problems with logical constraints. *SIAM Journal on Optimization*, 31(3):2340–2367, 2021.

[2] Rui Gao, Liyan Xie, Yao Xie, and Huan Xu. Robust hypothesis testing using Wasserstein uncertainty sets. In *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.